

Xylem Pipeline Relay

Move water uphill like a tree, faster and cleaner than rival teams.

Win condition: Most water delivered with speed, low spill, and strong explanation.

THE SCIENCE YOU ARE USING

Capillary action. Water climbs narrow spaces on its own. Water molecules stick to a surface (adhesion) and to each other (cohesion), so a thin column is pulled upward against gravity. Narrower channels and more wettable materials pull water higher and faster. Trees use this in their xylem.

Material and geometry. A felt strip, a cotton wick, and a folded paper towel each move water at a different rate. Path length, slope, and contact area all matter. Your team picks the material and the route to deliver the most water to a target cup.

HOW TO BUILD AND RUN IT

- 01 Test three wick materials.** Dip paper towel, cotton, and felt into colored water. Watch which climbs fastest and highest.
- 02 Pick a material and route.** Choose your wick and lay out a path from the source cup up over a block to the target cup.
- 03 Run the first transfer.** Start the wick and time how much water reaches the target in the time limit. Catch spills in the tray.
- 04 Reduce spill and length.** Trim the path, improve contact, and re-run. Less spill and a shorter path usually deliver more.
- 05 Final delivery round.** Run the best design for score. Water delivered, speed, and low spill all count.

Safety: Work over a tray and wipe spills immediately so the floor stays dry. No standing on chairs or stacked furniture to build height. Use washable food coloring; it can stain skin and clothes, so wear gloves if you do not want stained hands. No goggles are needed for moving water.

MATERIALS

Paper towels	shared (about 1 roll)
Cotton string	3 per team (12 in wicks)
Felt strips	3 per team (16 total)
Cups	2 per team (8)
Food coloring	shared (washable)
Trays	1 per team (4)
Elevation blocks	1 per team (4)
Rulers	1 per team (4)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

Wick we chose (paper towel / cotton string / felt) and why: _____

The ONE thing we changed for the redesign (path length, slope, contact, or material): _____

Water delivered: baseline _____ **after redesign** _____ **spill** _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	100